Supersingular poles and signed residues of prime-level Fricke Hauptmoduln

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Abstract

For $N \in \{2, 3, 5, 7, 13\}$, let $T_N = q^{-1} + O(q)$ be the normalized Hauptmodul for $\Gamma_0(N)^+$. For every prime $p \geq 5$ with $p \nmid N$, we determine the poles and residues of the rational function representing $T_N|U_p$ modulo p. Its poles are exactly the supersingular values with effective ramification index one. If t_i is such a value, its residue multiplied by $\prod_{k \neq i}(t_i - t_k)^{(p+1)/e_k}$ equals $-\left(\frac{-N}{p}\right)$, where the product includes all distinct supersingular values and e_k is the corresponding ramification index. The proof combines a Kaneko–Zagier derivative argument with Baker's supersingular congruence and an eta-character calculation. The latter gives $\big(\Delta(\tau)/\Delta(\ell\tau)\big)^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$ at every supersingular point of $X_0(\ell)$ for every prime $\ell \neq p$. The ramification indices also admit an interpretation as node thicknesses in the Deligne–Rapoport Fricke quotient. Exact computations provide finite certificates for all admissible pairs with $p \leq 211$.

Keywords: supersingular elliptic curves; Fricke groups; Hauptmoduln; Hecke operators; eta quotients; congruences.

1 Introduction

The reduction of U_p applied to a modular function can encode the supersingular locus as a divisor of poles. At level one, this relation is developed by de Shalit (1994) and Kaneko and Zagier (1998, §10); Baker (1998) proves the supersingular congruence that determines the sign in the level-one residue formula. Betina and Lecouturier (2018) obtain corresponding formulas for Legendre modular polynomials. We establish a uniform signed formula on the five prime-level Fricke curves for which $X_0(N)$ itself has genus zero.

The residues contain information beyond the support of the supersingular divisor. Since the reduced function vanishes at the cusp and has only simple poles, its residues determine the entire function. An explicit residue formula therefore reconstructs the reduced U_p image from supersingular values and their weights, and detects vanishing when no contributing value remains. The level-one theory provides the model for this reconstruction; the Fricke quotient introduces additional ramification, which must enter both the pole criterion and the product.

The distinction between ramified and unramified supersingular values is essential. A ramified value is removed from the pole divisor, but remains in the residue product with an exponent determined by its ramification index. The algebraic proof makes this distinction through the vanishing order of a quotient involving the Hasse invariant and the derivative of the Hauptmodul. Section 7 explains the same indices as thicknesses of nodes in an integral modular model.

The sign calculation also has a formulation independent of genus zero. Ono (2026, Theorem 1.2 and Remark 4) proves that the supersingular value of $\Delta(\tau)/\Delta(\ell\tau)$ raised to $(p^2-1)/12$ is one, and exhibits examples in which the half-exponent $(p^2-1)/24$ gives -1. Theorem 1.3 determines this remaining sign: it is $\left(\frac{\ell}{p}\right)$ for every prime $\ell \neq p$, at every supersingular point of $X_0(\ell)$. Thus

the eta/Frobenius identity applies without a genus-zero hypothesis, whereas its use in the explicit residue formula requires the coordinates fixed below.

1.1 Notation and normalization

Throughout the pole and residue results, $N \in \mathcal{N} := \{2, 3, 5, 7, 13\}$ and $p \geq 5$ is prime with $p \neq N$. Put

$$a_N = \frac{24}{N-1}, \qquad b_N = N^{12/(N-1)}, \qquad u_N = \left(\frac{\eta(\tau)}{\eta(N\tau)}\right)^{a_N}, \qquad T_N = u_N + a_N + b_N/u_N.$$

The pairs (a_N, b_N) are $(24, 4096), (12, 729), (6, 125), (4, 49), (2, 13)$. The Fricke involution sends u_N to b_N/u_N, and $u_N = q^{-1} - a_N + O(q)$, so $T_N = q^{-1} + O(q)$ has zero constant term. These are the McKay–Thompson series of classes $2A, 3A, 5A, 7A, 13A$ (Conway and Norton, 1979); the proofs below use their modular properties.

We write $T = T_N$, $u = u_N$, $q = e^{2\pi i\tau}$, $\theta = q\, d/dq$, $k = (p^2-1)/24$, $A = \overline{E}_{p-1}$ and $B = \overline{E}_{p+1}$. Thus A is the Hasse invariant, with q-expansion 1, and B has the q-expansion of E_2 modulo p. Put $\Delta = \eta^{24}$ and $\Delta_N = \Delta(\tau)\Delta(N\tau)$. For $f = \sum c_n q^n$, define $f|U_p = \sum c_{pn} q^n$. A bar denotes reduction modulo p. The finite fields are $\mathbb{F}_p$, $\mathbb{F}_{p^2}$ and $\overline{\mathbb{F}}_p$, and $\left(\frac{a}{p}\right)$ is the Legendre symbol. When a modular form is evaluated at an elliptic curve, an invariant differential is understood; all weight-zero ratios are independent of its choice.

The functions u and T remain coordinates on $X_0(N)_{\mathbb{F}_p}$ and $X_0(N)^+_{\mathbb{F}_p}$ respectively. Indeed, the eta quotient is a modular unit with divisor the difference of the two cusp sections, with unit leading coefficients at both cusps away from N. It therefore gives the integral genus-zero coordinate; the quotient by $u \mapsto b_N/u$ has coordinate T and a single cusp. This also fixes the normalization used in every product below.

1.2 Weights

Let $t_1, \ldots, t_s \in \overline{\mathbb{F}}_p$ be the distinct supersingular values of T. For a point $x = (E, C)$ of $X_0(N)_{\overline{\mathbb{F}}_p}$ over t_i put

$$m_x = |\mathrm{Aut}(E, C)/\{\pm 1\}|, \qquad e_i = m_x\, |\mathrm{Stab}_{\langle w_N\rangle}(x)|.$$

Call t_i *thin* if $e_i = 1$ and put $D_{\mathrm{thin}}(T) = \prod_{e_i=1}(T - t_i)$. For thin t_i let

$$r_i = \mathrm{Res}_{T=t_i} R_{N,p}(T), \qquad W_i = \prod_{k\neq i} (t_i - t_k)^{(p+1)/e_k}.$$

Section 3 shows that e_i does not depend on the representative of the Fricke orbit, that it is the effective (tame) inertia of the quotient map at x (not the order of a group containing ± 1), and that every exponent $(p+1)/e_k$ is an integer.

1.3 Main results

Theorem 1.1. *Let $N \in \{2, 3, 5, 7, 13\}$ and let $p \geq 5$ be a prime with $p \neq N$.*

(a) ***(Rationality.)*** *There is a unique $R_{N,p} \in \mathbb{F}_p(T)$ with $\overline{T|U_p} = R_{N,p}(\overline{T})$. It vanishes at $T = \infty$.*

(b) ***(Pole support.)*** *$R_{N,p} = P/D_{\mathrm{thin}}$ with $P \in \mathbb{F}_p[T]$, $\deg P < \deg D_{\mathrm{thin}}$ and $\gcd(P, D_{\mathrm{thin}}) = 1$. Every pole is simple and has a nonzero residue. In particular $\mathrm{den}_{\mathrm{red}} R_{N,p} = D_{\mathrm{thin}}$.*

(c) ***(Residue as a Hasse derivative.)*** *If t_i is thin and x lies over t_i, then*

$$r_i = \frac{144\,(\theta T)(x)^{p+1}}{B(x)^2}, \qquad \theta = q\frac{d}{dq},$$

a ratio of two mod-p modular forms of weight $2(p+1)$, evaluated at x.

(d) ***(Product.)*** *For thin t_i, with $x = (E, C)$ over t_i and $E' = E/C$,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \left(\frac{N}{p}\right) c(E)\, c(E'), \qquad c(E) := \frac{B(E,\omega)^{(p-1)/2}}{\Delta(E,\omega)^k}.$$

(e) ***(Signed weighted residue formula.)*** *For every thin t_i,*

$$\boxed{r_i W_i = -\left(\frac{-N}{p}\right)} \qquad \textit{equivalently} \qquad r_i = -\left(\frac{-N}{p}\right) \prod_{k \neq i} (t_i - t_k)^{-(p+1)/e_k}.$$

In particular $r_i \in \mathbb{F}_p^\times$ and $W_i \in \mathbb{F}_p^\times$, also when $t_i \notin \mathbb{F}_p$.

(f) ***(Hecke form.)*** *With $\Psi_{N,p}(X,Y) = (X^p - Y)(X - Y^p) + pH_{N,p}(X,Y)$ as in Proposition 2.1, $\overline{H}_{N,p}(t^p, t) = \left(\frac{-N}{p}\right) W_i^{-1}$ if $t = t_i$ is thin, and $\overline{H}_{N,p}(t^p, t) = 0$ for every other $t \in \mathbb{F}_{p^2}$.*

Corollary 1.2 (reconstruction)**.** *The supersingular values and their indices determine the entire reduced function:*

$$R_{N,p}(T) = -\left(\frac{-N}{p}\right) \sum_{e_i = 1} \frac{1}{W_i (T - t_i)}. \tag{1.1}$$

Equivalently, if $D_{\text{thin}} = \prod_f f$ is its factorization into monic irreducibles over $\mathbb{F}_p$, then

$$R_{N,p}(T) = \sum_{f \mid D_{\text{thin}}} c_f \frac{f'(T)}{f(T)}, \qquad c_f = -\left(\frac{-N}{p}\right) W_i^{-1} \in \mathbb{F}_p^\times \quad \textit{for any root } t_i \textit{ of } f. \tag{1.2}$$

Both sums are empty when $D_{\text{thin}} = 1$.

Proof. Subtracting the principal parts leaves a polynomial, which is zero because the function and each summand vanish at infinity. Frobenius-conjugate poles have conjugate residues; the residues lie in $\mathbb{F}_p$ by Theorem 1.1(e), so they are equal on each irreducible factor. The identity $f'/f = \sum_{f(t_i)=0} (T - t_i)^{-1}$ gives the second formula. □

Theorem 1.3 (Frobenius on the eta quotient)**.** *Let $\ell \neq p$ be prime. At every supersingular point x of $X_0(\ell)_{\overline{\mathbb{F}}_p}$,*

$$\left(\frac{\Delta(\tau)}{\Delta(\ell\tau)}\right)^{(p^2-1)/24}(x) = \left(\frac{\ell}{p}\right).$$

For $\ell = N \in \mathcal{N}$ this reads $u_N(x)^{(p^2-1)/a_N} = \left(\frac{N}{p}\right)$.

Corollary 1.4 (a square root of Baker's congruence)**.** *The value $c(E)$ is the same for all supersingular $E/\overline{\mathbb{F}}_p$. Call it c_p. Then $c_p \in \mathbb{F}_p$ and $c_p^2 = -\left(\frac{-1}{p}\right)$; equivalently*

$$E_{p+1}^{(p-1)/2} \equiv c_p\, \Delta^{(p^2-1)/24} \pmod{(p, E_{p-1})}.$$

Conjecture 1.5 (numerical evaluation)**.** $c_p = \left(\frac{6}{p}\right) \cdot \left(\frac{p-1}{2}\right)! \bmod p$.

1.4 Consequences

The reconstruction in Corollary 1.2 gives concrete formulas once the supersingular values and their effective indices are known. The connection to Nakaya's squarefree Fricke supersingular polynomials is a change of coordinate: with the normalizations in Nakaya (2019, §4),

$$\mathcal{S}_{N,p}(T) := \prod_i (T - t_i) = ss_p^{(N*)}(T + c_N), \qquad \begin{array}{c|ccccc} N & 2 & 3 & 5 & 7 & 13 \\ \hline c_N & 104 & 42 & 16 & 9 & 0 \end{array} \tag{1.3}$$

All constants are reduced modulo p; $\mathcal{S}_{N,p}$ is distinct from the weighted polynomial S of Proposition 3.4. Thus $D_{\mathrm{thin}} = \mathcal{S}_{N,p}/\prod_{e_i>1}(T-t_i)$, while the omitted values still occur in the weights W_i. The exceptional values can also be specified by a polynomial independent of p. Let h_N be the elliptic-value polynomial in Table 2. Lemma 3.2 and Theorem 1.1(b) give

$$D_{\mathrm{thin}}(T) = \frac{\mathcal{S}_{N,p}(T)}{\gcd(\mathcal{S}_{N,p}(T), \overline{h}_N(T))}, \tag{1.4}$$

where the gcd is monic. Thus the denominator is obtained from Nakaya's supersingular polynomial by removing its common factors with the reduction of a fixed characteristic-zero elliptic-value polynomial.

At level two, put $\varepsilon_p = (1 - \left(\frac{-1}{p}\right))/2$ and $\nu_p = (1 - \left(\frac{-2}{p}\right))/2$. The level-two indices described in Section 7 give

$$D_{2,p}(T) = \frac{ss_p^{(2*)}(T+104)}{(T+104)^{\varepsilon_p}(T-152)^{\nu_p}}. \tag{1.5}$$

The two removed values correspond to 0 and 256 in Nakaya's coordinate; they are distinct for $p \geq 5$.

Corollary 1.6 (level-two degree and vanishing)**.** *For every prime $p \geq 5$,*

$$\deg D_{2,p} = \frac{p - 4 + \left(\frac{-1}{p}\right) + 2\left(\frac{-2}{p}\right)}{8}. \tag{1.6}$$

In particular $T_{2A}|U_p \equiv 0 \pmod{p}$ if and only if $p = 5$ or 7.

Proof. The two possible thick values have indices four and two, and supersingularity indicators ε_p and ν_p, respectively. Corollary 3.5 therefore gives $\deg D_{2,p} = (p-1)/8 - \varepsilon_p/4 - \nu_p/2$, which is (1.6). This degree is zero at $p = 5, 7$ and is at least $(p-7)/8 > 0$ for $p \geq 11$. Theorem 1.1(b) proves the last assertion. □

Chen et al. (2019, Lemma 4.8 and Table 4.3) establish p-adic annihilation of T_{2A} at $p = 5, 7$. The corollary specifies the exact first-step modulo-p vanishing criterion. Table 1 illustrates the nonzero cases; at $p = 17$, for example, $R_{2,17} = D'_{2,17}/D_{2,17}$ and both residues are one.

p	$D_{2,p}(T)$	$P_{2,p}(T)$
11	$T - 5$	1
13	$T - 3$	-2
17	$T^2 - T - 8$	$2T - 1$
29	$T^3 - 12T^2 + 12T - 6$	$4T^2 - 5T - 8$

Table 1: Level-two examples, with $R_{2,p} = P_{2,p}/D_{2,p}$. Coefficients use the unique integer representatives in $(-p/2, p/2)$. These entries are included in the certificates of Section 8.

1.5 Relation to earlier work

The derivative argument in Section 4 follows the level-one method presented by Kaneko and Zagier (1998, §10), who trace it to Koike's work on Atkin's congruences (Koike, 1973). The period formula of de Shalit (1994) and the congruence of Baker (1998) supply the level-one precedents for the residue product and its sign; Betina and Lecouturier (2018) treat the corresponding Legendre modular polynomials. The isogeny transformation of B is due to Robert (1980, Théorème B, p. 106; §3.3, Lemma 7 and (19), p. 124); we include a q-expansion proof to fix the differential normalization. Baker's congruence supplies the level-one factor $-\left(\frac{-1}{p}\right)$.

Fricke supersingular polynomials are studied by Sakai (2011, 2014), Nakaya (2019) and Morton (2023). These papers treat orthogonal-polynomial representations, factorization and resultant identities. Formula (1.3) identifies the supersingular values used here with that literature; the present residue formula additionally assigns a nonzero coefficient to each thin pole and retains the ramified values in its weighted product.

Jorgenson et al. (2019, §4.2, Table 1 and Appendix C) determine the elliptic values of normalized Fricke Hauptmoduln and their associated class fields. Their j_N equals our T_N. Table 2 records their polynomials at the five levels considered here. Their Schwarzian equation detects the same local ramification indices that enter our Hasse-derivative calculation; (1.4) expresses its elliptic-value locus as the locus of possible supersingular pole cancellations. The signed residue formula concerns the remaining thin values.

Duncan and Swisher (2026, Proposition 3.1 and §4) use Deligne's supersingular expansion of $pJ|U_p$ and modular functions of prime-power level to characterize the exponents in the order of the Monster. Their sharp valuation bounds at the exceptional level-one invariants are closely related to the ramification mechanism considered here. Our calculation determines the signed residues of the reduced U_p images on the five fixed Fricke curves.

Half-exponent discriminant identities have an earlier precedent in Landweber (1988, Theorem 3, p. 72; §4). For a supersingular curve $y^2 = 4\prod_{a=1}^{3}(x - e_a)$ he proves

$$\Delta^{(p^2-1)/24} = (e_a - e_b)^{(p^2-1)/4} \quad (a \neq b).$$

Theorem 1.3 instead compares discriminants along a prime-degree isogeny, with its differential normalization built into a weight-zero function, and identifies the resulting sign by an eta character. Ono (2026, Theorem 1.2 and Remark 4) proves the squared isogeny-ratio identity and records examples with nontrivial half-exponent sign. His work develops the supersingular congruences arising in the description of the divided beta family by Behrens (2009, Theorem 1.2) and the explicit calculations of Larson (2020, §5.4, Conjecture 5.17). These connections motivate the discriminant identity; no new statement about the Adams–Novikov spectral sequence is needed here.

Our main result combines the signed isogeny-ratio identity with the Hasse-derivative calculation to determine the full residue products on the five Fricke curves. We make no claim that the exponent $(p^2 - 1)/24$ itself is new, or that the isogeny transformation of E_{p+1} originates here.

2 Standard inputs and the residue criterion

We use the q-expansion principle and Kodaira–Spencer isomorphism for prime-to-p level structures (Gross, 1990; Katz, 1973; Katz and Mazur, 1985). On a fine moduli cover, the Hasse invariant has simple zeros precisely at the supersingular points, and Kodaira–Spencer identifies a weight-two form f with $f\,dq/q$. In particular, θT represents dT. We record the statements used repeatedly.

Standard fact I1. The form A has q-expansion 1 and simple supersingular zeros on the moduli stack. The q-expansion principle holds at prime-to-p level. For $p \geq 5$ the automorphism groups relevant to $X_0(N)$ and its Fricke quotient are tame. Coarse quotients commute with reduction. The supersingular values of T lie in $\mathbb{F}_{p^2}$.

The last assertion follows by choosing a model of a supersingular curve with p^2-Frobenius $[-p]$ (Lemma 5.8); every cyclic subgroup of order N is then rational as a subgroup scheme.

Standard fact I2 (Baker's congruence)**.** For every $p \geq 5$,

$$E_{p+1}^{p-1} \equiv -\left(\frac{-1}{p}\right)\Delta^{(p^2-1)/12} \pmod{(p, E_{p-1})}.$$

This is Baker (1998, Theorem 1.4), with the normalized Eisenstein series used here.

Standard fact I3. For every prime $\ell \neq p$, the supersingular ℓ-isogeny graph over $\overline{\mathbb{F}}_p$ is connected (Mestre, 1986; Pizer, 1990).

Proposition 2.1. *Let Δ_p be the set of integral matrices of determinant p with lower-left entry divisible by N. There is a unique $\Psi_{N,p} \in \mathbb{Z}[X,Y]$ with*

$$\Psi_{N,p}(X, T(\tau)) = (X - T(p\tau)) \prod_{b=0}^{p-1} \Big(X - T\big(\tfrac{\tau+b}{p}\big)\Big).$$

It is monic of degree $p+1$ in each variable, symmetric, and satisfies $\Psi_{N,p} \equiv (X^p - Y)(X - Y^p) \pmod p$.

Proof. The matrices $\mathrm{diag}(p,1)$ and $\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$, $0 \le b < p$, represent $\Gamma_0(N)\backslash\Delta_p$. Conjugation by $W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ maps $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to $\begin{pmatrix} d & -c/N \\ -Nb & a \end{pmatrix}$, so it preserves Δ_p. Hence the elementary symmetric functions of the roots are $\Gamma_0(N)^+$-invariant, holomorphic on the upper half-plane, with a pole only at the single cusp. Their coefficients initially lie in $\mathbb{Z}[\zeta_p]((q^{1/p}))$. Translation of τ removes fractional powers, and cyclotomic Galois automorphisms permute the roots, so the expansions lie in $\mathbb{Z}((q))$. Subtracting integral powers of $T = q^{-1} + O(q)$ then puts them in $\mathbb{Z}[T]$. Modulo $1 - \zeta_p$ all $T((\tau+b)/p)$ coincide with $T(q^{1/p})$, whose p-th power is $T(q)$, and $T(p\tau) \equiv T^p$; this gives the congruence. For irreducibility, the reduction of $\Gamma_0(N)$ modulo p is $\mathrm{SL}_2(\mathbb{F}_p)$, by the Chinese remainder theorem and the surjectivity of reduction for $\mathrm{SL}_2(\mathbb{Z})$. It acts transitively on the $p+1$ cyclic subgroups of order p. The $p+1$ roots displayed above are distinct: their leading terms are q^{-p} and $\zeta_p^{-b} q^{-1/p}$ for $0 \le b < p$. They therefore form one orbit, so $\Psi_{N,p}(X,T)$ is irreducible over $\mathbb{C}(T)$. The pole order of the jth elementary symmetric function is at most $p + (j-1)/p \le p+1$; the constant coefficient has pole order $p+1$. Thus the polynomial has degree $p+1$ in Y as well as in X. For symmetry, $\mathrm{diag}(1,p) \in \Delta_p$ gives $\Psi_{N,p}(T(\tau), T(p\tau)) = 0$ by evaluating the defining relation at $p\tau$. Irreducibility and Gauss's lemma therefore identify the reversed integral primitive polynomial with the original up to sign. Hence $\Psi_{N,p}(Y,X) = c\,\Psi_{N,p}(X,Y)$ with $c = \pm 1$. Reducing modulo p, where both polynomials equal the symmetric $(X^p - Y)(X - Y^p)$, forces $c = 1$. □

Define $H = H_{N,p} \in \mathbb{Z}[X,Y]$ by $\Psi_{N,p} = (X^p - Y)(X - Y^p) + pH$; H is symmetric. Put

$$\psi = \frac{T(q)^p - T(q^p)}{p} \in \mathbb{Z}((q)),$$

which is integral because $T(q^p) \equiv T(q)^p \pmod p$.

Proposition 2.2 (residue criterion)**.** *In $\mathbb{F}_p((q))$,*

$$\overline{\psi} = \frac{\overline{H}(T^p, T)}{T^{p^2} - T}, \qquad \overline{T|U_p} = \overline{Q}(T) + \overline{\psi},$$

where $\overline{Q} \in \mathbb{F}_p[T]$ is defined by $-[X^p]\Psi_{N,p} = Y^p + pQ(Y)$. Hence $R_{N,p} = \overline{Q}(T) + \overline{H}(T^p, T)/(T^{p^2} - T)$ exists and is unique. It vanishes at ∞, its poles are simple and lie in $\mathbb{F}_{p^2}$, and for $t \in \mathbb{F}_{p^2}$

$$\mathrm{Res}_{T=t}\, R_{N,p} = -\overline{H}(t^p, t).$$

Proof. Substitute $X = T(q^p) = T^p - p\psi$ and $Y = T$ into $\Psi_{N,p} = 0$. Since $X - Y^p = -p\psi$, division by p and reduction give $-\overline{\psi}\,(T^{p^2} - T) + \overline{H}(T^p, T) = 0$. The sum of the roots of $\Psi_{N,p}(X,T)$ is $T(q^p) + p\,(T|U_p) = T^p + pQ(T)$, so $T|U_p = Q(T) + \psi$. The polynomial $T^{p^2} - T$ is squarefree, its roots are exactly $\mathbb{F}_{p^2}$, and its derivative is -1. $R_{N,p}$ vanishes at ∞ because $T|U_p \in q\mathbb{Z}[[q]]$ (the constant term of T is 0). Uniqueness holds because $\overline{T}$ is transcendental over $\mathbb{F}_p$. □

Corollary 2.3. *For every $t \in \mathbb{F}_{p^2}$, $\operatorname{Res}_t R_{N,p} \in \mathbb{F}_p$ and $\operatorname{Res}_{t^p} R_{N,p} = \operatorname{Res}_t R_{N,p}$.*

Proof. $(\operatorname{Res}_t R)^p = \operatorname{Res}_{t^p} R = -\overline{H}(t^{p^2}, t^p) = -\overline{H}(t, t^p) = -\overline{H}(t^p, t) = \operatorname{Res}_t R$, using $t^{p^2} = t$, the $\mathbb{F}_p$-rationality of R and the symmetry of H. □

Proposition 2.2 gives Theorem 1.1(a) and the a priori bound $\operatorname{den}_{\mathrm{red}} R_{N,p} \mid T^{p^2} - T$ used by the certificates in Section 8.

3 Weights and the weighted supersingular polynomial

Lemma 3.1 (effective inertia)**.** *Let $x = (E, C)$ be a non-cuspidal point of $X_0(N)_{\overline{\mathbb{F}}_p}$, supersingular or ordinary. Let G_x be the group of automorphisms of (E, C) together with the isomorphisms $(E, C) \to w_N(E, C) = (E/C, E[N]/C)$ when $w_N(x) = x$. It is a group once $w_N^2(E, C) \cong (E, C)$ is identified via $[N]$.*

(i) *$|G_x/\{\pm 1\}| = m_x |\operatorname{Stab}_{\langle w_N \rangle}(x)| = e_x$. The value is the same at x and at $w_N(x)$.*

(ii) *$\{\pm 1\}$ is exactly the kernel of the action of G_x on the deformation space of x. Since $e_x \in \{1, 2, 3, 4, 6\}$ is prime to p, $G_x/\{\pm 1\}$ is cyclic and acts faithfully on the tangent line. Hence, for a local parameter z of a fine moduli cover at a point over x, $T - t = z^{e_x} \cdot (\text{unit})$.*

(iii) *If x is supersingular, $(p + 1)/e_x \in \mathbb{Z}$.*

Proof. (i) If $w_N(x) = x$, the isomorphisms $(E, C) \to w_N(E, C)$ form a torsor under $\operatorname{Aut}(E, C)$, so $|G_x| = 2|\operatorname{Aut}(E, C)|$. Conjugation by the isogeny $E \to E/C$ identifies $\operatorname{Aut}(E, C)$ with $\operatorname{Aut}(E/C, E[N]/C)$, so $m_x = m_{w_N(x)}$. (ii) $[-1]$ acts trivially on deformations. Suppose $g \in G_x$, $g \neq \pm 1$, acted trivially. Then g would extend to the universal deformation. Its generic fibre has endomorphism ring $\mathbb{Z}$, and g corresponds either to a nontrivial automorphism or to an endomorphism of degree N, which is not a square. That is impossible. The same argument works at ordinary points. A finite group of order prime to p acting faithfully on $k[[z]]$ is cyclic and acts faithfully on $(z)/(z^2)$, so the invariant ring is $k[[z^e]]$. (iii) $e = 3$ or 6 forces $m_x = 3$, so $j = 0$ is supersingular and $p \equiv 2 \pmod 3$; $e = 4$ forces $m_x = 2$, so $j = 1728$ is supersingular and $p \equiv 3 \pmod 4$. For $p \geq 5$, $m_x \in \{1, 2, 3\}$. □

For explicit calculations, m_x can be read off from the multiplicity μ of $u(x)$ as a root of $P_N(u) - j\, u^N$: $m_x = 3/\mu$ over $j = 0$, $m_x = 2/\mu$ over $j = 1728$, and $m_x = 1$ (with $\mu = 1$) otherwise. These are ramification indices of the coarse map $X_0(N) \to X(1)$. The Fricke stabilizer has order two exactly when $u(x)^2 = b_N$.

Lemma 3.2 (elliptic values and specialization)**.** *Let $h_N(T) = \prod_\tau (T - T_N(\tau))$, where τ runs over the inequivalent elliptic points of $\Gamma_0(N)^+$. For the five levels under consideration, h_N is given in Table 2. For every prime $p \geq 5$ with $p \neq N$, its reduction is squarefree, and its roots in $\overline{\mathbb{F}}_p$ are exactly the values with effective index $e > 1$. Consequently,*

$$\gcd(\mathcal{S}_{N,p}, \overline{h}_N) = \prod_{e_i > 1} (T - t_i).$$

Proof. The characteristic-zero values and elliptic orders are those of Jorgenson et al. (2019, §4.2, Table 1 and Appendix C), in the same normalization $T_N = q^{-1} + O(q)$. Their elliptic orders are the effective indices of Lemma 3.1 in characteristic zero. We justify the specialization of their support. A nontrivial effective stabilizer arises either from an automorphism of (E, C) or from a Fricke fixed point. The former can occur only over $j = 0$ or 1728. Since $p \geq 5$ and $p \nmid N$, the corresponding elliptic curve and its automorphisms lift to characteristic zero, and the prime-to-p cyclic subgroup C lifts uniquely over a strict henselian base. An automorphism preserving C on

the special fibre therefore preserves its lift. Conversely, these prime-to-p automorphisms persist under reduction. Fricke fixed points are given by $u^2 = b_N$, which is finite étale away from $2N$ and has the same fixed-point locus under specialization. This accounts for all values with $e > 1$ in either characteristic. Directly from the displayed polynomials, the discriminants of h_N, in the order $N = 2, 3, 5, 7, 13$, are

$$2^{16}, \qquad 2^4 \cdot 3^6, \qquad 2^{12} \cdot 5^3, \qquad 2^4 \cdot 3^6 \cdot 7^2, \qquad 2^{12} \cdot 3^6 \cdot 13.$$

Thus distinct elliptic values do not collide at any admissible prime. The roots of $\overline{h}_N$ are precisely the values with $e > 1$; intersecting this set with the supersingular locus proves the gcd formula. □

N	$h_N(T)$	Effective indices
2	$(T+104)(T-152)$	4, 2
3	$(T+42)(T-66)$	6, 2
5	$(T+16)(T^2-12T-464)$	2, 2
7	$(T+10)(T+9)(T-18)$	2, 3, 2
13	$(T+4)(T+3)(T^2-4T-48)$	2, 3, 2

Table 2: Elliptic-value polynomials from Jorgenson et al. (2019). The indices are listed in the order of the displayed factors; an index beside a quadratic factor applies to both roots. At admissible primes, the supersingular roots of $\overline{h}_N$ are exactly the thick values removed in (1.4).

Lemma 3.3. *As mod-p forms on $\Gamma_0(N)$: $A|_{p-1}W_N = \left(\frac{N}{p}\right) A$, $\Delta_N|_{24}W_N = \Delta_N$ and $\theta T|_2 W_N = \theta T$.*

Proof. For a level-one form f of weight k, $f|_k W_N = N^{k/2} f(N\tau)$. The form $E_{p-1}(N\tau)$ has q-expansion $\equiv 1 \equiv E_{p-1}(\tau)$, so it equals A modulo p by Fact I1(i); and $N^{(p-1)/2} \equiv \left(\frac{N}{p}\right)$. The other two identities hold in characteristic 0: $\Delta(-1/(N\tau))\Delta(-1/\tau) = N^{12}\tau^{24}\Delta(N\tau)\Delta(\tau)$, and T is w_N-invariant. □

Proposition 3.4 (weighted supersingular polynomial)**.** *Let $S(T) = \prod_{k=1}^{s} (T - t_k)^{(p+1)/e_k}$. Then $S \in \mathbb{F}_p[T]$ and*

$$S(\overline{T}) = \Delta_N^{-k} = q^{-(N+1)k} \prod_{n \geq 1} (1-q^n)^{-24k} (1-q^{Nn})^{-24k} \quad \text{in } \mathbb{F}_p((q)).$$

Equivalently, $S = A^{p+1}/\Delta_N^k$ as functions on $X_0(N)^+_{\mathbb{F}_p}$. In particular $\sum_k e_k^{-1} = (N+1)(p-1)/24$.

Proof. $F := A^{p+1}/\Delta_N^k$ is a ratio of mod-p forms of weight p^2-1. By Lemma 3.3 it is w_N-invariant, since $\left(\frac{N}{p}\right)^{p+1} = 1$. Δ_N does not vanish on $Y_0(N)$, so F is regular off the cusp of $X_0(N)^+$. Its q-expansion there is $\Delta_N^{-k} = q^{-(N+1)k}(1+O(q))$, because A has q-expansion 1. Hence F is a monic polynomial in T of degree $(N+1)k$. By Fact I1(ii) and Lemma 3.1(ii), F vanishes exactly at the supersingular t_k, to order $(p+1)/e_k$ in $T - t_k$. Comparing degrees gives the mass identity. □

Corollary 3.5. *The degree of the reduced denominator is*

$$\deg D_{\text{thin}} = \frac{(N+1)(p-1)}{24} - \sum_{e_i > 1} \frac{1}{e_i}.$$

In particular $R_{N,p} = 0$ exactly when no thin supersingular value exists.

Proof. Separate the thin and non-thin terms in the mass identity of Proposition 3.4; use Theorem 1.1(b) for the last assertion. □

4 The Kaneko–Zagier step

The following is the level-N version of Kaneko and Zagier (1998, §10, (35)–(38)), who credit the method to Koike.

Lemma 4.1. *In $\mathbb{F}_p((q))$, with $d/dT = (\theta T)^{-1}\theta$,*

$$\frac{d\overline{\psi}}{dT} = T^{p-1} - (\theta T)^{p-1}, \qquad (\theta T)^{p-1} = g(\overline{T}), \quad g := \frac{(\theta T)^{p-1}}{A^2} \in \mathbb{F}_p(T).$$

Proof. Over $\mathbb{Z}$, $\theta\psi = T^{p-1}\theta T - (\theta T)(q^p)$, and $(\theta T)(q^p) \equiv (\theta T)^p$. Divide by $\theta T = -q^{-1} + \cdots$, which is invertible. The function g is a ratio of mod-p forms of weight $2(p-1)$, it is w_N-invariant by Lemma 3.3, and it has q-expansion $(\theta T)^{p-1}$ because $A \equiv 1$. □

Lemma 4.2 (local structure of g)**.** *Let $t \in \mathbb{F}_{p^2}$ be a non-cuspidal point with effective inertia e_t, and put $\varepsilon_t = 1$ if t is supersingular and 0 otherwise. Then*

$$\operatorname{ord}_{T=t} g = \frac{(p-1)(e_t-1) - 2\varepsilon_t}{e_t}.$$

For thin t and x over t, let $a = A/(\theta T)^{(p-1)/2}$, a function on $X_0(N)$ near x. Then a has a simple zero at x,

$$a'(t) := \frac{da}{dT}(x) = \frac{(B/12)(x)}{(\theta T)(x)^{(p+1)/2}} \neq 0, \qquad g = a'(t)^{-2}(T-t)^{-2}(1 + O(T-t)).$$

Proof. By Fact I1(iv) and Lemma 3.1(ii), θT vanishes to order $e_t - 1$ in a stack parameter z, and A to order ε_t; also $T - t \sim z^{e_t}$. This gives the order formula. For thin t, the map $X_0(N) \to X_0(N)^+$ is étale at x and $\theta T(x) \neq 0$. In characteristic 0, for forms F and M of equal weight, $\theta(F/M) = (\partial F \cdot M - F \cdot \partial M)/M^2$, where $\partial = \theta - \frac{k}{12}E_2$ is the Serre derivative. Apply this with $F = E_{p-1}$ and $M = (\theta T)^{(p-1)/2}$ and reduce modulo p; at x we have $A(x) = 0$. The form ∂E_{p-1} has weight $p+1$ and q-expansion $\equiv \theta(1) + \frac{1}{12}E_2 \equiv \frac{1}{12}E_{p+1}$, using the Kummer congruence $2(p+1)/B_{p+1} \equiv 24$ and $\sigma_p \equiv \sigma_1$. So $\partial A = B/12$ as mod-p forms. Finally $da/dT = \theta a/\theta T$. The value $B(x) \neq 0$ because $B/12$ is the derivative of A at its simple zero. □

Proof of Theorem 1.1(b)–(c). Let $t \in \mathbb{F}_{p^2}$. By Proposition 2.2, near t we have $\overline{\psi} = -\overline{H}(t^p, t)(T-t)^{-1} + O(1)$, because $T^{p^2} - T = -(T-t)(1 + O(T-t))$. So the coefficient of $(T-t)^{-2}$ in $d\overline{\psi}/dT$ is $\overline{H}(t^p, t)$. By Lemma 4.1 it is also $-[g]_{-2}$. Hence

$$\operatorname{Res}_t R_{N,p} = -\overline{H}(t^p, t) = [g]_{-2}.$$

By Lemma 4.2, $\operatorname{ord}_t g \geq 0$ unless t is thin. For $p \geq 5$ and $e_t \geq 2$ we have $(p-1)(e_t-1) - 2 \geq 0$. At thin t, $[g]_{-2} = a'(t)^{-2} = 144(\theta T)(x)^{p+1}/B(x)^2 \neq 0$. Every pole is simple by Proposition 2.2, so $R_{N,p} = P/D_{\text{thin}}$ with $P(t_i) \neq 0$ at every thin t_i. Since $R_{N,p}(\infty) = 0$, $\deg P < \deg D_{\text{thin}}$. □

Lemma 4.3 (Robert's isogeny identity)**.** *Let N be any prime $\neq p$ (not only $N \in \mathcal{N}$). Let $x = (E, C)$ be a supersingular point of $X_0(N)$, $\pi : E \to E' = E/C$, and $\omega = \pi^*\omega'$. Then $B(E', \omega') = N\,B(E, \omega)$, $A(E', \omega') = A(E, \omega)$ and $\Delta(N\tau)(x) = N^{-12}\Delta(E', \omega')$.*

Proof. The isogeny identity is Robert (1980, Théorème B; §3.3, Lemma 7); see also Baker (1998, Lemma 2.2). We verify the normalization by q-expansions. For a level-one form f of weight w, $f(N\tau)$ at (E, C, ω) equals $N^{-w}f(E', \omega')$. Check this on $(\mathrm{Tate}(q), \mu_N)$, where $E' \cong \mathrm{Tate}(q^N)$ and $\pi^*\omega'_{\text{can}} = N\omega_{\text{can}}$. The form $E_2(\tau) - NE_2(N\tau)$ is a holomorphic weight-2 form on $\Gamma_0(N)$ with p-integral coefficients. Modulo p, the forms $B(\tau) - NB(N\tau)$ and $A \cdot (E_2 - NE_2(N\tau))$ have weight $p+1$ and the same q-expansion, so they are equal. At a supersingular point this gives $B(E, \omega) = N \cdot N^{-(p+1)}B(E', \omega') = N^{-1}B(E', \omega')$, because $N^{p-1} = 1$. The statement about A follows in the same way from $A(N\tau) = A(\tau)$. □

Proposition 4.4 (weighted residue product)**.** *For thin t_i and $x = (E, C)$ over t_i,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \beta(E)\Big(\frac{\Delta(\tau)}{\Delta(N\tau)}\Big)^k(x) = \left(\frac{N}{p}\right) c(E)c(E'), \qquad \beta(E) := \frac{B(E,\omega)^{p-1}}{\Delta(E,\omega)^{2k}} = c(E)^2.$$

Proof. Near x, Proposition 3.4 gives $S(T) = A^{p+1}/\Delta_N^k = a^{p+1}\,Q^k$ with $Q := (\theta T)^{12}/\Delta_N$, a function that is regular and nonzero at x. Hence $W_i = [S/(T-t_i)^{p+1}]_{T=t_i} = a'(t_i)^{p+1}Q(x)^k$. With $r_i = a'(t_i)^{-2}$ this gives

$$r_i W_i = a'(t_i)^{p-1}Q(x)^k = \Big(\frac{B}{12}\Big)^{p-1}(\theta T)^{-\frac{p^2-1}{2}} \cdot \frac{(\theta T)^{\frac{p^2-1}{2}}}{\Delta_N^k} = \frac{B^{p-1}}{\Delta_N^k}(x).$$

Write $\Delta_N^k = \Delta^{2k}(\Delta(N\tau)/\Delta)^k$ for the second equality. For the third, Lemma 4.3 gives $\Delta_N(x) = N^{-12}\Delta(E,\omega)\Delta(E',\omega')$ with $N^{12k} = N^{(p^2-1)/2} = 1$, and $B(E)^{(p-1)/2} = N^{-(p-1)/2}B(E')^{(p-1)/2} = \left(\frac{N}{p}\right)B(E')^{(p-1)/2}$. □

5 The eta quotient at supersingular points

Throughout this section $\ell \neq p$ is a prime, $M := 576\ell$, $k := (p^2-1)/24$, and

$$D_\ell(\tau) := \frac{\Delta(\tau)}{\Delta(\ell\tau)}, \qquad G(\tau) := \frac{\eta(24\ell\tau)}{\eta(24\tau)}.$$

Then $G = q^{\ell-1}\prod_{n\geq 1}(1-q^{24\ell n})/(1-q^{24n}) \in q^{\ell-1}\mathbb{Z}[[q]]$ and $G^{24} = D_\ell(24\tau)^{-1}$. We prove Theorem 1.3: $D_\ell(x)^k = \left(\frac{\ell}{p}\right)$ at every supersingular point x of $X_0(\ell)_{\overline{\mathbb{F}}_p}$. The argument has four steps: (i) the multiplier of G on $\Gamma_0(M)$; (ii) the moduli interpretation of D_ℓ, of G, and of the diamond operators; (iii) the Frobenius action on a lift of x to $X_1(M)$; (iv) the descent.

5.1 The multiplier of G

We use the closed form of the eta multiplier (Rademacher; Petersson; see Knopp, 1970, Ch. 4, §4.1, and Apostol, 1990, Thm. 3.4 for the Dedekind-sum form): for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ with $c > 0$,

$$\eta(\gamma\tau) = \nu(\gamma)\,(c\tau+d)^{1/2}\,\eta(\tau), \qquad \nu(\gamma) = \begin{cases} \left(\frac{d}{c}\right) e\Big(\dfrac{(a+d)c - bd(c^2-1) - 3c}{24}\Big), & c \text{ odd}, \\ \left(\frac{c}{d}\right) e\Big(\dfrac{(a+d)c - bd(c^2-1) + 3d - 3 - 3cd}{24}\Big), & c \text{ even}, \end{cases} \tag{5.1}$$

where $e(x) = e^{2\pi i x}$, $(c\tau+d)^{1/2}$ is the principal branch, and the symbols are Kronecker symbols (so $\left(\frac{c}{d}\right)$ is defined for $d < 0$, with $\left(\frac{c}{-1}\right) = \operatorname{sgn} c$).

Lemma 5.1 (multiplier of G)**.** *Let $\chi_\ell(d) := \left(\frac{\ell}{d}\right)$ (Kronecker symbol). For every $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(M)$,*

$$G(\gamma\tau) = \chi_\ell(d)\,G(\tau).$$

The function $d \mapsto \chi_\ell(d)$ on $(\mathbb{Z}/M)^\times$ is the Kronecker character of $\mathbb{Q}(\sqrt{\ell})$: it is a Dirichlet character of conductor ℓ ($\ell \equiv 1 \bmod 4$), 4ℓ ($\ell \equiv 3 \bmod 4$) or 8 ($\ell = 2$), each dividing M; it is even and quadratic, and $\chi_\ell(p) = \left(\frac{\ell}{p}\right)$ is the Legendre symbol.

Proof. If $c = 0$ then $\gamma = \pm \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $G(\tau + b) = G(\tau)$ because G has an integral q-expansion, and $\chi_\ell(\pm 1) = 1$. If $c < 0$ replace γ by $-\gamma$, which acts in the same way on $\mathfrak{H}$ and has $\chi_\ell(-d) = \chi_\ell(d)$ since $\left(\frac{\ell}{-1}\right) = 1$. So let $c > 0$. Put $c_1 = c/24$, $c_2 = c/(24\ell)$, $b_1 = 24b$, $b_2 = 24\ell b$ and $\gamma_i = \begin{pmatrix} a & b_i \\ c_i & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then $24\gamma\tau = \gamma_1(24\tau)$ and $24\ell\gamma\tau = \gamma_2(24\ell\tau)$, and both automorphy factors equal $(c\tau + d)^{1/2}$, so $G(\gamma\tau) = (\nu(\gamma_2)/\nu(\gamma_1))G(\tau)$. Since $576\ell \mid c$, both c_1 and c_2 are divisible by 24, in particular even, so the second line of (5.1) applies to both. In the quotient of the exponentials every term is a multiple of 24: $(a + d)(c_2 - c_1)$ and $3d(c_2 - c_1)$ because $24 \mid c_i$, and $d(b_2(c_2^2 - 1) - b_1(c_1^2 - 1))$ because $24 \mid b_i$. Hence the exponentials cancel and

$$\frac{\nu(\gamma_2)}{\nu(\gamma_1)} = \left(\frac{c_2}{d}\right)\left(\frac{c_1}{d}\right) = \left(\frac{c_1 c_2}{d}\right) = \left(\frac{\ell c_2^2}{d}\right) = \left(\frac{\ell}{d}\right)\left(\frac{c_2}{d}\right)^2 = \left(\frac{\ell}{d}\right),$$

using $c_1 = \ell c_2$ and $\gcd(c_2, d) = 1$. The statements about χ_ℓ are quadratic reciprocity: for $d > 0$ odd, $\left(\frac{\ell}{d}\right) = \left(\frac{d}{\ell}\right)$ if $\ell \equiv 1 \pmod 4$, $\left(\frac{\ell}{d}\right) = \left(\frac{-1}{d}\right)\left(\frac{d}{\ell}\right)$ if $\ell \equiv 3 \pmod 4$, and $\left(\frac{2}{d}\right) = \chi_8(d)$; in each case this is the Kronecker character of discriminant ℓ, 4ℓ, 8 respectively, which is even because $\mathbb{Q}(\sqrt{\ell})$ is real. For a prime $p \nmid M$, $\chi_\ell(p)$ is the Legendre symbol. □

Remark 5.2. The multiplier lemma is also an instance of the Gordon–Hughes–Newman eta-quotient transformation law (Ono, 2004, Theorem 1.64). For $y = \eta(\tau)/\eta(\ell\tau)$, the corresponding formula requires $24\ell \mid c$ and $24 \mid b$; the latter hypothesis cannot in general be omitted.

5.2 Moduli conventions

The μ_M-model. For $M \geq 5$ and over $\mathbb{Z}[1/M]$ let $Y_1(M)$ be the fine, affine moduli scheme of pairs (E, ι) with $\iota : \mu_M \hookrightarrow E$ a closed immersion of group schemes, and $X_1(M) \supset Y_1(M)$ its smooth proper compactification (Deligne and Rapoport, 1973, IV–V); (Gross, 1990, §2); (Katz, 1976, §1). The Tate curve $\mathrm{Tate}(q) = \mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{Z}[1/M]((q))$ with the canonical embedding $\mu_M \subset \mathbb{G}_m$ gives a section ∞ of $X_1(M)$ over $\mathbb{Z}[1/M]$ whose formal completion is $\operatorname{Spf} \mathbb{Z}[1/M][[q]]$. The complex points are $\Gamma_1(M)\backslash\mathfrak{H}^*$ via $\tau \mapsto x_\tau := (E_\tau, \iota_\tau)$, $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, $\iota_\tau(e^{2\pi i/M}) = 1/M$; under $E_\tau \cong \mathbb{C}^\times/q^{\mathbb{Z}}$, $z \mapsto e^{2\pi i z}$, this is the canonical embedding, so the q-expansion of a function at ∞ is its Fourier expansion in $q = e^{2\pi i\tau}$. The diamond operators are the $\mathbb{Z}[1/M]$-automorphisms

$$\langle d\rangle(E, \iota) = (E, \iota \circ [d]), \qquad d \in (\mathbb{Z}/M)^\times.$$

Convention. We write $\langle d\rangle$ for this automorphism of the scheme and $f \circ \langle d\rangle$ for the pull-back of a function; no inverse occurs anywhere below.

Lemma 5.3 (analytic diamond operators)**.** *For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(M)$ one has $x_{\gamma\tau} = \langle d\rangle x_\tau$. Consequently, for a function f on $X_1(M)_{\mathbb{C}}$ with $f(\gamma\tau) = \chi(d)f(\tau)$ on $\Gamma_0(M)$,*

$$f \circ \langle d\rangle = \chi(d)\, f.$$

Proof. Multiplication by $c\tau + d$ maps $\mathbb{Z} + \mathbb{Z}\gamma\tau$ onto $\mathbb{Z} + \mathbb{Z}\tau$ (it sends $\gamma\tau \mapsto a\tau + b$ and $1 \mapsto c\tau + d$, and $\det\gamma = 1$), hence is an isomorphism $E_{\gamma\tau} \to E_\tau$. It sends $\iota_{\gamma\tau}(\zeta_M) = 1/M$ to $(c\tau + d)/M \equiv d/M = \iota_\tau(\zeta_M^d)$ because $M \mid c$. So $(E_{\gamma\tau}, \iota_{\gamma\tau}) \cong (E_\tau, \iota_\tau \circ [d])$. Then $(f \circ \langle d\rangle)(x_\tau) = f(x_{\gamma\tau}) = f(\gamma\tau) = \chi(d)f(x_\tau)$. □

Lemma 5.4 (q-expansion descent)**.** *Let $f \in \mathbb{C}(X_1(M))$ have poles supported at the cusps and Fourier expansion at ∞ in $\mathbb{Q}((q))$. Then $f \in \mathbb{Q}(X_1(M))$ for the μ_M-model.*

Proof. The reduced cusp divisor is defined over $\mathbb{Q}$. Choose a multiple D of it bounding the poles of f. The space $H^0(X_1(M), \mathcal{O}(D))$ is defined over $\mathbb{Q}$, and its base change to $\mathbb{C}$ contains f. Expansion at the rational cusp, in the rational parameter q, gives a linear map defined over $\mathbb{Q}$. Finitely many Laurent coefficients give an injective map on this finite-dimensional space: otherwise a nonzero section could vanish to arbitrarily high order at ∞. In a rational basis this injective coefficient map is a rational matrix. The specified coefficients of f are rational, so its coordinates in that basis are rational. This proves descent from $\mathbb{C}$, including algebraicity. □

Remark 5.5 (the point convention). For the model classifying (E, P) with P of order M, the standard Tate cusp $(\mathrm{Tate}(q), \zeta_M)$ is defined over $\mathbb{Q}(\zeta_M)$ rather than $\mathbb{Q}$ (Conrad, n.d., Example 2.3). Rational Fourier coefficients do not by themselves give the same descent assertion on that model. We use only the μ_M-model, its rational Tate cusp and the diamond convention fixed above.

The function D_ℓ on $Y_0(\ell)$. For a pair (E, C) over a $\mathbb{Z}[1/\ell]$-scheme, with $\pi : E \to E' = E/C$ the quotient isogeny and ω' any nowhere-vanishing invariant differential on E', put

$$D_\ell(E, C) := \ell^{12}\, \frac{\Delta(E, \pi^*\omega')}{\Delta(E', \omega')} \in \mathcal{O}(S)^\times. \tag{5.2}$$

This is independent of ω' (both discriminants have weight 12), so it is an invertible function on $Y_0(\ell)_{\mathbb{Z}[1/\ell]}$. On the Tate curve with $C = \mu_\ell$, $E' = \mathrm{Tate}(q^\ell)$ via $t \mapsto t^\ell$, $\pi^*(dt'/t') = \ell\, dt/t$, so $\Delta(E, \pi^*\omega'_{\mathrm{can}}) = \ell^{-12}\Delta(q)$ and $\Delta(E', \omega'_{\mathrm{can}}) = \Delta(q^\ell)$: the q-expansion of (5.2) is $\Delta(q)/\Delta(q^\ell)$. Hence (5.2) is the algebraic incarnation of $\Delta(\tau)/\Delta(\ell\tau)$, with the value at a geometric point $x = (E, C)$ given by (5.2); this is the meaning of "$D_\ell(x)$" in Theorem 1.3 (with the differential normalization of Lemma 4.3).

Lemma 5.6 (the degeneracy map)**.** *The rule*

$$\delta(E, \iota) = \Big(E/\iota(\mu_{24}),\quad \iota(\mu_{24\ell})/\iota(\mu_{24})\Big)$$

defines a morphism $\delta : Y_1(M) \to Y_0(\ell)$ over $\mathbb{Z}[1/M]$ (here $\mu_{24} \subset \mu_{24\ell} \subset \mu_M$). On complex points $\delta(x_\tau) = (E_{24\tau}, \langle 1/\ell\rangle)$, i.e. δ is $\tau \mapsto 24\tau$, and

$$G^{24} = \delta^*(D_\ell^{-1}) \quad \text{in } \mathbb{Q}(X_1(M)).$$

For every $x \in Y_0(\ell)(\overline{\mathbb{F}}_p)$ there is $\tilde{x} \in Y_1(M)(\overline{\mathbb{F}}_p)$ with $\delta(\tilde{x}) = x$.

Proof. The quotient by the finite étale subgroup $\iota(\mu_{24})$ and the image of $\iota(\mu_{24\ell})$ (cyclic of order ℓ) are defined over $\mathbb{Z}[1/M]$. Analytically, multiplication by 24 induces $E_\tau \to \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\,24\tau) = E_{24\tau}$ with kernel $\langle 1/24\rangle = \iota_\tau(\mu_{24})$, and maps $\iota_\tau(\mu_{24\ell}) = \langle 1/(24\ell)\rangle$ onto $\langle 1/\ell\rangle$. The function D_ℓ at $(E_{\tau'}, \langle 1/\ell\rangle)$ is $\Delta(\tau')/\Delta(\ell\tau')$, so $(\delta^* D_\ell)(x_\tau) = \Delta(24\tau)/\Delta(24\ell\tau) = G(\tau)^{-24}$; Lemma 5.4 applies to G, whose poles are cuspidal and whose Fourier coefficients are integral; thus $G \in \mathbb{Q}(X_1(M))$. Equality on complex points now gives the asserted identity in that rational function field. For surjectivity on $\overline{\mathbb{F}}_p$-points, choose a cyclic subgroup $K \subset E$ of order 24, requiring $K[\ell] \neq C$ when $\ell = 2$ or 3. There is such a choice because the ℓ-torsion has $\ell + 1$ lines, and each line extends to a cyclic subgroup of the required prime-power order. Let $q_K : E \to E_1 = E/K$ and let $\widehat{q}_K : E_1 \to E$ be its dual, with kernel K_1. Elementary divisors of prime-to-p torsion show that $H = \widehat{q}_K^{-1}(C)$ is cyclic of order 24ℓ. For clarity, at a prime $r \mid 24$, choose torsion coordinates in which K is the first coordinate line. The dual quotient has elementary divisors $(1, r^{v_r(24)})$. If $r = \ell$, the condition $C \neq K[r]$ makes its preimage cyclic of order $r^{v_r(24)+1}$; at $r \neq \ell$ it is the cyclic kernel. When $\ell \nmid 24$, the additional ℓ-part is a line and the same conclusion follows. Extend H to a cyclic subgroup $Z \subset E_1$ of order $M = 24^2\ell$. Its unique subgroup of order 24 is K_1. Choose an isomorphism $\iota : \mu_M \xrightarrow{\sim} Z$ over $\overline{\mathbb{F}}_p$. The dual quotient identifies E_1/K_1 with E, and $\iota(\mu_{24\ell})/K_1$ with C. Thus $\delta(E_1, \iota) = x$. □

Proposition 5.7 (descent and integrality of G)**.** *$G \in \mathbb{Q}(X_1(M))$, and G is an invertible regular function on $Y_1(M)_{\mathbb{Z}_{(p)}}$. Its reduction $\overline{G}$ is an invertible regular function on $Y_1(M)_{\mathbb{F}_p}$, defined over $\mathbb{F}_p$, and*

$$\overline{G} \circ \langle d \rangle = \chi_\ell(d)\, \overline{G} \qquad (d \in (\mathbb{Z}/M)^\times), \qquad \overline{G}^{\,24} = \delta^*(\overline{D}_\ell^{-1}).$$

Proof. G is holomorphic and non-vanishing on $\mathfrak{H}$, invariant under $\Gamma_1(M)$ (Lemma 5.1, $\chi_\ell(1) = 1$) and meromorphic at the cusps, so $G \in \mathbb{C}(X_1(M))$; by Lemma 5.4 and the integral q-expansion, $G \in \mathbb{Q}(X_1(M))$. By Lemma 5.6, $F := \delta^*(D_\ell^{-1})$ lies in $\mathcal{O}(Y_1(M)_{\mathbb{Z}[1/M]})^\times$ and $G^{24} = F$. The ring $R := \mathcal{O}(Y_1(M)_{\mathbb{Z}_{(p)}})$ is a normal domain (smooth over $\mathbb{Z}_{(p)}$, flat with integral generic fibre), and $G \in \mathrm{Frac}(R)$ satisfies the monic equation $T^{24} - F = 0$ with $F \in R^\times$; hence $G \in R$, and $G^{-1} = G^{23}F^{-1} \in R$. The identity $G \circ \langle d \rangle = \chi_\ell(d)G$ holds in $\mathbb{C}(X_1(M))$ by Lemmas 5.1 and 5.3, hence in R ($\langle d \rangle$ is an automorphism of $Y_1(M)$ over $\mathbb{Z}[1/M]$), hence after reduction. Reduction of $G^{24} = F$ gives the last identity. □

5.3 Frobenius on a lift

Lemma 5.8 (Deuring models)**.** *Let $E_1/\overline{\mathbb{F}}_p$ be supersingular, $p \geq 5$. Then E_1 has a model over $\mathbb{F}_{p^2}$ whose p^2-power Frobenius endomorphism is $[-p]$. For such a model every finite subgroup of order prime to p is $\mathbb{F}_{p^2}$-rational, and E_1/K again has Frobenius $[-p]$.*

Proof. For any model $E/\mathbb{F}_{p^2}$ of a supersingular curve, the p-power Frobenius $F : E \to E^{(p)}$ and the Verschiebung $V : E^{(p)} \to E$ are both purely inseparable of degree p (the Hasse invariant vanishes), so $F^{(p)} : E^{(p)} \to E^{(p^2)} = E$ and V differ by an automorphism u of E: $F^{(p)} = u \circ V$, whence $\pi_{p^2} = F^{(p)} \circ F = u \circ V \circ F = u \circ [p]$. Thus $\pi_{p^2} = p\,u$ with $u \in \mathrm{Aut}(E)$. If $j \neq 0, 1728$ then $u = \pm 1$ and the quadratic twist changes the sign; if $j = 0$ (resp. 1728) the curve $y^2 = x^3 + 1$ (resp. $y^2 = x^3 + x$) over $\mathbb{F}_p$ has trace 0, so $\pi_{p^2} = \pi_p^2 = [-p]$ (Deuring, 1941; Waterhouse, 1969); see also Ono (2026, Lemma 4.3). Since $[-p]$ is central, Galois preserves every subgroup, and the quotient isogeny ϕ satisfies $\pi'\phi = \phi\pi = \phi[-p] = [-p]\phi$, so $\pi' = [-p]$. □

Proof of Theorem 1.3. Let $x = (E, C) \in Y_0(\ell)(\overline{\mathbb{F}}_p)$ be supersingular and choose, by Lemma 5.6, a lift $\tilde{x} = (E_1, \iota) \in Y_1(M)(\overline{\mathbb{F}}_p)$ with $\delta(\tilde{x}) = x$. The curve E_1 is isogenous to E, hence supersingular, so by Lemma 5.8 we may replace (E_1, ι) by an isomorphic pair (same geometric point $\tilde{x}$, same value $\overline{G}(\tilde{x})$) in which E_1 is defined over $\mathbb{F}_{p^2}$ with $\pi_{E_1} = [-p]$. The embedding $\iota : \mu_M \to E_1$ is defined over $\overline{\mathbb{F}}_p$; nothing requires it to be $\mathbb{F}_{p^2}$-rational.

Galois action on the point. Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_{p^2})$ be $u \mapsto u^{p^2}$. Since $Y_1(M)$ is a fine moduli scheme, σ acts on $Y_1(M)(\overline{\mathbb{F}}_p)$ by $\sigma(E_1, \iota) = (E_1^\sigma, \iota^\sigma) = (E_1, \sigma_{E_1} \circ \iota \circ \sigma_{\mu_M}^{-1})$, where $\sigma_{\mu_M}(\zeta) = \zeta^{p^2}$ and σ_{E_1} is the action on $E_1(\overline{\mathbb{F}}_p)$ through the $\mathbb{F}_{p^2}$-structure, i.e. the Frobenius endomorphism $\pi_{E_1} = [-p]$. Hence

$$\iota^\sigma(\zeta) = [-p]\,\iota(\zeta^{p^{-2}}) = [-p][p^{-2}]\,\iota(\zeta) = \iota([-p^{-1}]\zeta), \qquad \text{i.e.} \qquad \sigma(\tilde{x}) = \langle -p^{-1} \rangle \tilde{x},$$

with $-p^{-1}$ read in $(\mathbb{Z}/M)^\times$. (In the point convention the same computation gives $\sigma(E_1, P) = (E_1, [-p]P) = \langle -p \rangle'(E_1, P)$; the final answer below is unchanged because χ_ℓ is even and quadratic, but we fix the μ_M-convention and use only the displayed formula.)

Galois action on the value. $\overline{G}$ is defined over $\mathbb{F}_p$ (Proposition 5.7), so $\overline{G}(\sigma\tilde{x}) = \sigma(\overline{G}(\tilde{x})) = \overline{G}(\tilde{x})^{p^2}$. Combining with the diamond identity of Proposition 5.7,

$$\overline{G}(\tilde{x})^{p^2} = \overline{G}(\langle -p^{-1} \rangle \tilde{x}) = \chi_\ell(-p^{-1})\,\overline{G}(\tilde{x}) = \chi_\ell(p)\,\overline{G}(\tilde{x}) = \left(\frac{\ell}{p}\right)\overline{G}(\tilde{x}),$$

because $\chi_\ell(-1) = 1$ and $\chi_\ell(p^{-1}) = \chi_\ell(p)^{-1} = \chi_\ell(p)$. Since $\overline{G}$ is invertible on $Y_1(M)_{\mathbb{F}_p}$, $\overline{G}(\tilde{x}) \neq 0$ and

$$\overline{G}(\tilde{x})^{p^2-1} = \left(\frac{\ell}{p}\right). \tag{5.3}$$

No coefficient-field factor enters: the only Galois action used is that of σ on the point $\tilde{x}$, and $\overline{G}$ is σ-invariant as a function.

Descent to $X_0(\ell)$. By Proposition 5.7, $\overline{G}(\tilde{x})^{24} = \overline{D}_\ell(\delta\tilde{x})^{-1} = D_\ell(x)^{-1}$. Therefore

$$D_\ell(x)^k = \overline{G}(\tilde{x})^{-24k} = \overline{G}(\tilde{x})^{-(p^2-1)} = \left(\frac{\ell}{p}\right)^{-1} = \left(\frac{\ell}{p}\right).$$

This holds at every geometric supersingular point, including those over $j = 0$ and $j = 1728$: the argument takes place on the fine moduli scheme $Y_1(M)$, and $D_\ell(x)$ is defined by (5.2) independently of automorphisms of (E, C). For $\ell = N \in \mathcal{N}$, $D_N = u_N^{N-1}$ and $(N-1)k = (p^2-1)/a_N$. □

Remark 5.9 (comparison with Ono). Ono (2026, Thm. 1.2) (v2, 3 July 2026) proves $R_\ell^{(p^2-1)/12} = 1$ at supersingular points for $R_\ell = \Delta(\ell\tau)/\Delta(\tau) = D_\ell^{-1}$, via $R_\ell(\tilde{P})^{(p^2-1)/12} = \kappa(E')/\kappa(E)$ (his Lemma 4.4) and $\kappa(E) = \Delta(E,\omega)^{(p^2-1)/12} = (-1)^{(p+1)/2}$ on Deuring models with $\mathbb{F}_{p^2}$-rational ω (his Thm. 1.3). His Remark 4 observes that the $(p^2-1)/24$ power can be -1 ($p = 11$, $\ell = 2$: $R_2 \in \{6, 7, 2\}$, fifth powers -1), that the values of $g_\ell = \eta(\ell z)/\eta(z)$ need not lie in $\mathbb{F}_{p^2}$ (they generate $\mathbb{F}_{11^4}$ there), and that a proof descending $\Delta^{1/24}$ or $\Delta^{1/12}$ to $\mathbb{F}_{p^2}$ must fail. Theorem 1.3 determines the sign: $R_\ell^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$, equivalently $R_\ell(\tilde{P})$ is a 24-th power in $\mathbb{F}_{p^2}^\times$ iff $\left(\frac{\ell}{p}\right) = 1$. This is consistent with Remark 4: $\left(\frac{2}{11}\right) = -1$, and (5.3) shows $\overline{G}(\tilde{x}) \in \mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$ exactly when $\left(\frac{\ell}{p}\right) = -1$. The proof above does not descend any root of Δ to $\mathbb{F}_{p^2}$: it uses that $\overline{G}$ is an $\mathbb{F}_p$-rational *function* on $Y_1(576\ell)$, and tracks its values through the diamond character. The isogeny-invariance of κ over $\mathbb{F}_{p^2}$ (for E_1, E_2 separably isogenous over $\mathbb{F}_{p^2}$, the values $(\left(\frac{-1}{p}\right)\Delta(E_i))^{(p^2-1)/12}$ coincide) is already in Baker (1998, §2), as a consequence of the Gross–Landweber congruence; Ono's Theorem 1.2 follows from it by his Lemma 4.4.

6 The signed formula and a level-one corollary

Proof of Theorem 1.1(e). By Proposition 4.4, $r_i W_i = \beta(E)\,(\Delta/\Delta(N\tau))^k(x)$. Fact I2 gives $\beta(E) = -\left(\frac{-1}{p}\right)$, and Theorem 1.3 gives $(\Delta/\Delta(N\tau))^k(x) = \left(\frac{N}{p}\right)$. So $r_i W_i = -\left(\frac{-1}{p}\right)\left(\frac{N}{p}\right) = -\left(\frac{-N}{p}\right)$. By Corollary 2.3, $r_i \in \mathbb{F}_p^\times$, and therefore $W_i = -\left(\frac{-N}{p}\right) r_i^{-1} \in \mathbb{F}_p^\times$. □

Proof of Theorem 1.1(f). By Proposition 2.2 and part (b), $\overline{H}(t^p, t) = -\operatorname{Res}_t R_{N,p}$. This is 0 unless t is thin, and equals $-r_i = \left(\frac{-N}{p}\right) W_i^{-1}$ if $t = t_i$ is thin. This fixes the sign: with $\Psi = (X^p - Y)(X - Y^p) + pH$, the residue is $-\overline{H}(t^p, t)$, not $+\overline{H}$. □

Proof of Corollary 1.4. Combining Lemma 4.3 with Theorem 1.3 gives, for every supersingular ℓ-isogeny $E \to E'$,

$$\frac{c(E')}{c(E)} = \left(\frac{\ell}{p}\right)(\Delta/\Delta(\ell\tau))^k(x) = 1.$$

By Fact I3, c is constant. Fact I2 gives $c^2 = \beta = -\left(\frac{-1}{p}\right)$. Since $-\left(\frac{-1}{p}\right)$ is -1 for $p \equiv 1 \pmod 4$ and $+1$ for $p \equiv 3 \pmod 4$, $c \in \mathbb{F}_p$ in both cases. For the congruence: the level-one mod-p form $B^{(p-1)/2} - c\,\Delta^{(p^2-1)/24}$ vanishes on the supersingular locus. The zeros of A there are simple on the stack, including at $j = 0, 1728$. Hence A divides it in the ring $\mathbb{F}_p[E_4, E_6]$ of level-one forms ($p \geq 5$). □

Remark 6.1. Given Lemma 4.3, Theorem 1.3 for all primes ℓ is equivalent to the constancy of c. It therefore follows from Theorem 1.3 for a single ℓ together with Fact I3.

Remark 6.2 (on Conjecture 1.5). By Wilson's theorem $(\frac{p-1}{2})!^2 \equiv (-1)^{(p+1)/2} = -\left(\frac{-1}{p}\right)$, so the conjecture is consistent with Corollary 1.4. Since $c(E)$ is constant, evaluation at one supersingular curve for each p would suffice. Choosing such a curve uniformly is less immediate than this reduction suggests: $j = 1728$ is supersingular only for $p \equiv 3 \pmod 4$, and $j = 0$ only for $p \equiv 2 \pmod 3$. In particular neither familiar CM model is supersingular when $p \equiv 1 \pmod{12}$. Moreover Baker's congruence fixes c_p^2 rather than choosing between its two square roots; an explicit evaluation of B is still needed. The factorial formula holds in the recorded computations for all primes $5 \le p \le 997$ and remains a conjecture here.

Remark 6.3 ($\mathbb{F}_p$-rationality at non-rational points). The two rationality assertions in Theorem 1.1(e) have different explanations. $r_i \in \mathbb{F}_p$ is elementary (Corollary 2.3: symmetry of H and Frobenius-equivariance of residues). $W_i \in \mathbb{F}_p$ is not formal: writing $W_i = \mathrm{Nm}_{\mathbb{F}_{p^2}/\mathbb{F}_p}(D'_{\mathrm{thin}}(t_i)) \cdot \prod_{e_k>1}(t_i - t_k)^{(p+1)/e_k}$, it amounts to the product $\prod_{e_k>1}(t_i - t_k)^{(p+1)/e_k}$ lying in $\mathbb{F}_p$. Equivalently, a product of power-residue characters (one of order e_k for each factor) must be trivial; this is not visible factor by factor. Here it follows only from part (e).

7 Geometric interpretation of the weights

The proof of Theorem 1.1 is algebraic. We now interpret its weights in the integral modular curve, retaining the geometric explanation of pole cancellation. Put $W = W(\overline{\mathbb{F}}_p)$ and let

$$\mathcal{B} = X_0(Np)^{\mathrm{DR}}_W, \qquad \mathcal{A} = \mathcal{B}/\langle w_N \rangle.$$

Here $\mathcal{B}$ denotes the coarse Deligne–Rapoport model and w_N is the partial Atkin–Lehner involution. A split node with completed local ring $W[[s,t]]/(st - p^e v)$, v a unit, has *thickness* e.

Proposition 7.1. *The special fibre of $\mathcal{A}$ has two components, each isomorphic to $X_0(N)^+_{\overline{\mathbb{F}}_p}$. They meet at the supersingular Fricke orbits. At the orbit of $x = (E, C_N)$ the node has thickness*

$$e_x = |\operatorname{Aut}(E, C_N)/\{\pm 1\}| \, |\operatorname{Stab}_{\langle w_N \rangle}(x)|.$$

Consequently $R_{N,p}$ has a pole at a supersingular value precisely when the corresponding node has thickness one.

Proof. Deligne–Rapoport's theorem applies with prime-to-p level N and the stabilizer of a line in $(\mathbb{Z}/N)^2$ as the auxiliary group (Deligne and Rapoport, 1973, V, 1.14–1.20; VI, Theorem 6.9, especially (6.9.1)). It gives two copies of $X_0(N)_{\overline{\mathbb{F}}_p}$ with Frobenius gluing, and the coarse completed local ring at x is

$$W[[s,t]]/(st - p^{m_x}), \qquad m_x = |\operatorname{Aut}(E, C_N)/\{\pm 1\}|.$$

The forgetful map $\mathcal{B} \to X_0(N)_W$ is finite: it is proper and has finite fibres on the generic fibre and on the two special-fibre components. Normality identifies $\mathcal{B}$ with the normalization in its generic function field. The generic partial Fricke map covers w_N on $X_0(N)_W$, hence extends uniquely by normalization. On the elliptic locus it is

$$(E, C_N, C_p) \longmapsto (E/C_N, E[N]/C_N, q_N(C_p)).$$

Since q_N is prime to p, it preserves the Frobenius and Verschiebung components and both branches at a node.

At a fixed node the stabilizer has order two and acts nontrivially on either coarse branch. Indeed its action on the smooth curve $X_0(N)$ is a nontrivial tame involution, whose derivative at a fixed point is -1. Taking the branch-preserving quotient therefore multiplies the thickness by this order, by Liu (2002, Proposition 10.3.48(b)–(c), pp. 526–528). At a non-fixed node the inertia is trivial and the thickness is unchanged. Invariants commute with reduction because 2 is

invertible in W. The quotient components are consequently $X_0(N)^{+}_{\mathbb{F}_p}$, and no node is smoothed, since no stabilizer exchanges branches. This proves the thickness formula. The last assertion is Theorem 1.1(b). □

Remark 7.2 (level two). For $N = 2$, the Fricke fixed points are $u = \pm 64$, with $T = -104, 152$. At $u = -64$ the reduced automorphism group has order two and the Fricke stabilizer has order two, giving thickness four. At $u = 64$ their orders are one and two, giving thickness two. All other supersingular values have thickness one. The two exceptional values are supersingular precisely when $\left(\frac{-1}{p}\right) = -1$ and $\left(\frac{-2}{p}\right) = -1$, respectively. Thus a thick value is absent from the pole divisor but remains in the weighted residue product. The same distinction applies at each of the five levels.

8 Finite certificates and computational evidence

The accompanying exact-arithmetic computations provide checks independent of the signed product formula. They are not used in the proof. We describe the certification bound so that the strength of each computation is explicit.

Proposition 8.1 (finite certificate)**.** *Let $L = p^2$, let $D \in \mathbb{F}_p[T]$ be a monic divisor of $T^L - T$, of degree d, and let $P \in \mathbb{F}_p[T]$ have degree less than d. If*

$$D(\overline{T})\,\overline{T|U_p} - P(\overline{T}) = O(q^{L-d+1}),$$

then $R_{N,p} = P/D$. For $D = 1$, take $P = 0$.

Proof. Proposition 2.2 implies that $R_{N,p} - P/D = C/(T^L - T)$ with $\deg C < L$. If $C \neq 0$, its q-order is $L - \deg C$, since $T = q^{-1} + O(q)$. Multiplication by $D(T)$ gives order $L - \deg C - d \leq L - d$, contradicting the hypothesis. Thus $C = 0$. □

The first computation reconstructs $R_{N,p}$ from its q-expansion, using $D = D_{\text{thin}}$ built independently from supersingular data. The coefficients of the numerator are recovered without the residue formula, and Proposition 8.1 then proves the reconstructed identity. The reduced denominator was also recovered without a candidate for $p \leq 37$. The Hecke polynomial was computed modulo p^2 for $p \leq 31$, checking symmetry, the Kronecker congruence, and the residue criterion at every element of $\mathbb{F}_{p^2}$.

N	certified pairs	thin values	non-rational thin values	range
2	45	530	310	$5 \leq p \leq 211$
3	45	716	456	$5 \leq p \leq 211$
5	44	1059	740	$5 \leq p \leq 211$, $p \neq 5$
7	44	1431	1116	$5 \leq p \leq 211$, $p \neq 7$
13	44	2510	2074	$5 \leq p \leq 211$, $p \neq 13$

Table 3: Certified reconstructions. Thin values are distinct geometric values of T_N.

All 222 admissible pairs with $p \leq 211$ passed the certificates, giving 6246 thin values, of which 4696 are non-rational. Four pairs have no thin values; the ratio set is empty in these cases. In every other pair it is a singleton equal to $-\left(\frac{-N}{p}\right)$. A second calculation, using the double-pole formula of Lemma 4.2, checks a further 960 pairs for $223 \leq p \leq 1499$. That calculation depends on the derivative lemma and is not a separate reconstruction from $T|U_p$.

The eta/Frobenius identity was also tested by a different route: normalized Vélu quotients and their discriminants, without eta functions or Hauptmoduln. For the 15 primes $\ell \leq 47$ and all admissible $5 \leq p \leq 199$, the calculation covers 647 pairs and 127541 cyclic-kernel checks on

chosen representatives of supersingular elliptic curves. These are not counts of distinct coarse points: automorphisms may identify kernels over $j = 0$ and 1728. Every kernel gave the asserted discriminant-ratio sign.

Data and software. The supplementary packages `Signed_Weighted_Residue_v1_0` and `Eta_Frobenius_Lemma_Audit_v1_0` contain Python code, per-pair JSON results and reproduction instructions (`python-flint` 0.9.0 and `mpmath`). The numerical ranges reported here are those recorded in these packages. The present revision changes the proofs, attribution and presentation; it does not assert a new full rerun of those ranges.

9 Further directions

The argument separates three inputs: the genus-zero coordinate and Kronecker congruence, the Hasse divisor and Kodaira–Spencer derivative, and the signed isogeny-ratio identity. Only the first two are specific to the pole theorem on the five curves. Theorem 1.3 applies to every prime isogeny degree different from p, without a genus-zero assumption. Extending the residue theorem to other Fricke curves requires an integral normalized coordinate and an explicit treatment of the cusp divisor; it is not asserted here.

The constant c_p of Corollary 1.4 gives a level-one formulation of the same sign information. Its factorial evaluation in Conjecture 1.5 remains open in this paper. The genus-zero moonshine identification supplies examples of the modular functions to which the theorem applies; the result itself is a statement about their reduction and supersingular divisors.

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